Problem

Let

$$n = 2^{19}3^{12}.$$

Let M denote the number of positive divisors of n^2 which are less than n but do not divide n.

Determine the number formed by the last two digits of M.

Source: **IOQM**

§1 Counting Divisors of n and n^2

Write

$$n = 2^{19}3^{12}.$$

Set

$$a = 12, \qquad b = 19.$$

Then

$$n = 3^a 2^b, \qquad n^2 = 3^{2a}2^{2b}.$$

The total number of divisors of n^2 is

$$(2a+1)(2b+1).$$

Since

$$(2a+1)(2b+1)$$

is odd, the divisor n lies exactly in the middle of the divisor list of n^2.

Indeed, if $d \mid n^2$, then the complementary divisor is

$$\frac{n^2}{d}.$$

For every divisor $d < n$, its complement satisfies

$$\frac{n^2}{d} > n,$$

and vice versa.

Hence the number of divisors of n^2 that are less than or equal to n is

$$\frac{(2a+1)(2b+1)+1}{2}.$$

Substituting $a = 12$ and $b = 19$,

$$\frac{(25)(39)+1}{2} = \frac{976}{2} = 488.$$

Equivalently,

$$488 = 2ab + a + b + 1.$$

§2 Counting Divisors of n

The number of positive divisors of n is

$$(a+1)(b+1).$$

Since every divisor of n is automatically less than or equal to n, the number of divisors of n not exceeding n is

$$(a+1)(b+1).$$

Substituting the values of a and b,

$$(a+1)(b+1) = 13 \cdot 20 = 260.$$

§3 Extracting the Required Count

We seek divisors of n^2 which

- are less than n,
- do not divide n.

Thus

$$M = \big(\text{divisors of } n^2 \text{ not exceeding } n\big) - (\text{divisors of } n)\,.$$

Therefore

$$\begin{aligned} M &= (2ab + a + b + 1) - (a+1)(b+1) \\ &= 2ab + a + b + 1 - (ab + a + b + 1) \\ &= ab. \end{aligned}$$

Hence

$$M = 12 \cdot 19 = 228.$$

§4 The Final Conclusion

The last two digits of M are

$$\boxed{28}.$$